

Hello! 🖐️



Pieter De Vis



Sam Hunt



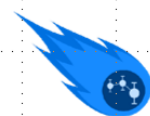
Maddie Stedman



Samantha Malone



Joe Riordan



Today's Exercises



[comet-toolkit.org/user-
guide/training/vhroda](https://comet-toolkit.org/user-guide/training/vhroda)



Today's Agenda

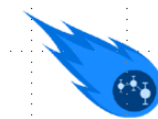


1. Presentation: CoMet Toolkit Introduction

- ☐ *What is CoMet?*
- ☐ *Uncertainty 101*
- ☐ *Tools Intro*

2. Exercises

- ☐  **punpy** basics with in-situ type data
- ☐  **obsarray** basics with EO type data



Today's Exercises



comet-toolkit.org/user-guide/training/vhroda



The CoMet Toolkit – Uncertainties made easy

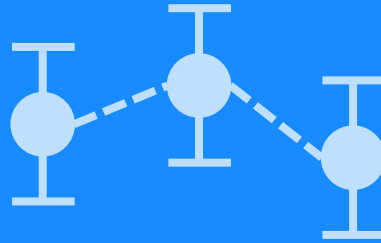
Pieter De Vis, Sam Hunt, Maddie Stedman, Samantha Malone, Joe
Riordan
National Physical Laboratory

VHRODA hands-on tutorial - 19/11/2025

CoMet Toolkit



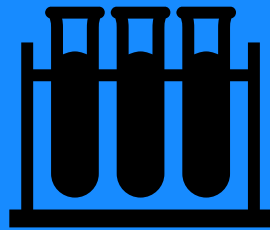
Python
Tools



Uncertainty
Handling



Open
Source



Tested

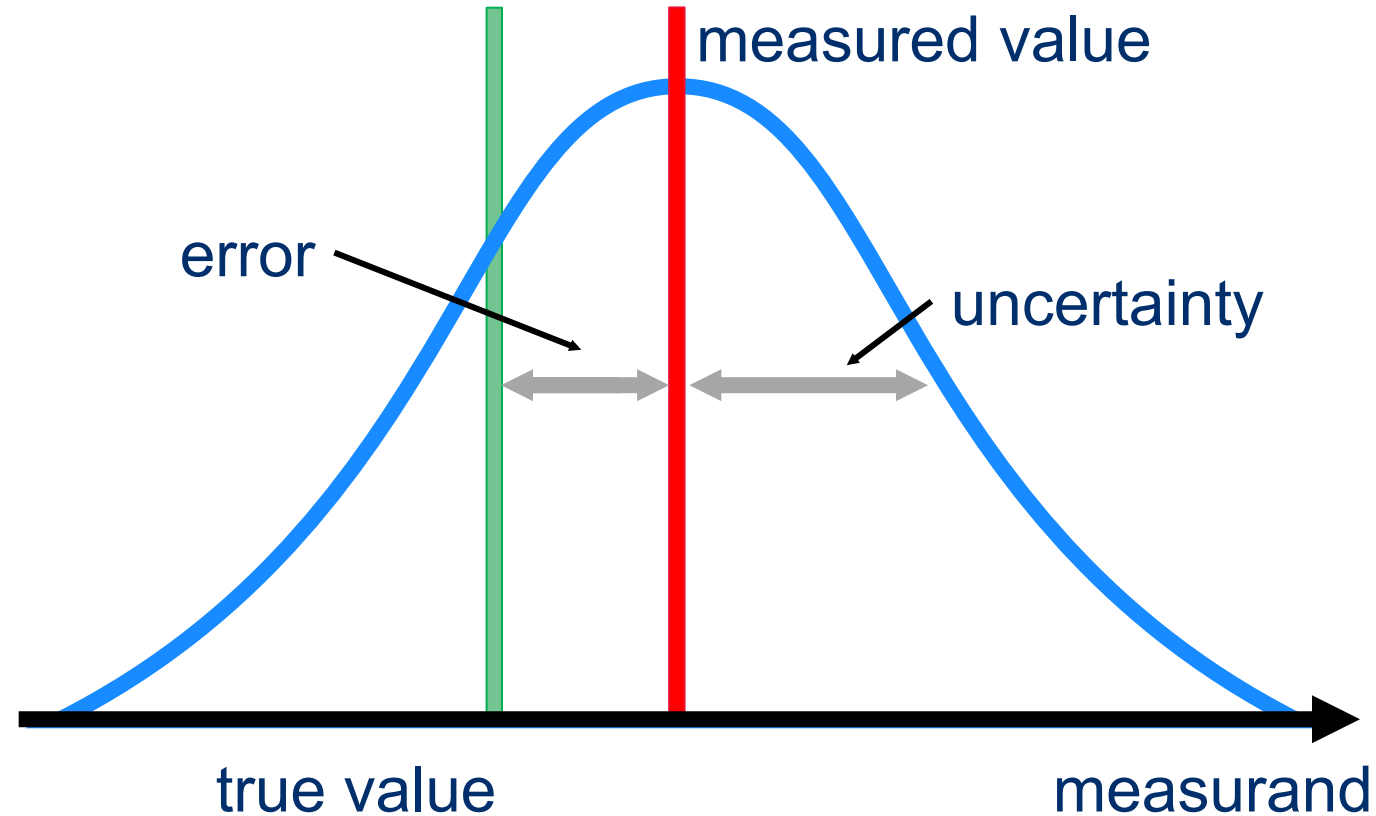


Applied

IDEAS-QA4EO



Uncertainties 101





Error Correlation



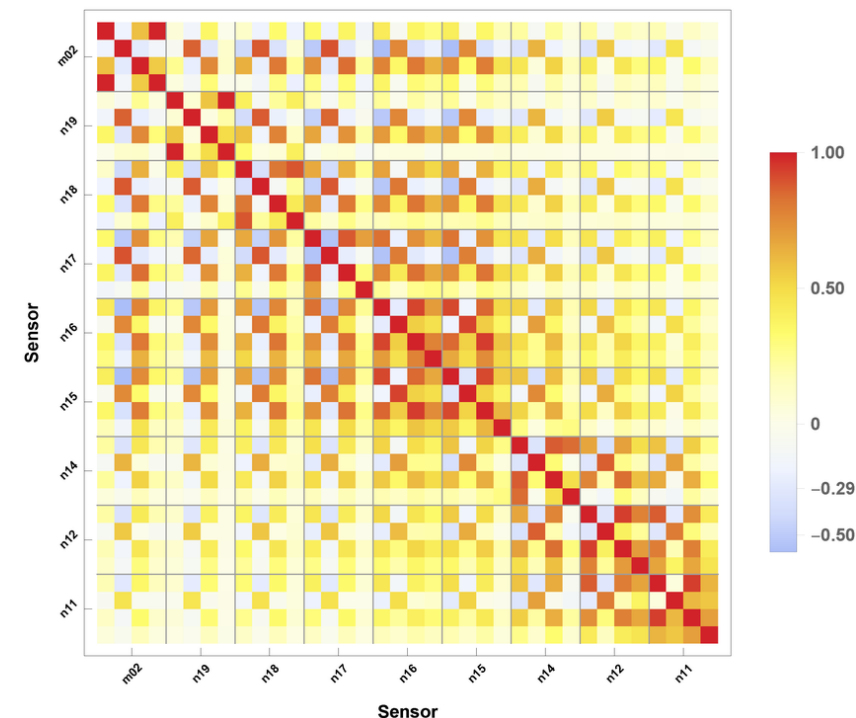
What is Error Correlation?

Errors in a dataset are **not always independent** — e.g., when errors in pixels, bands instruments, or time steps are systematically related.



Why does it Matter?

- ☐ Bias uncertainties persist – averaging doesn't help!
- ☐ Band ratios might be off – affecting retrieval uncertainty
- ☐ Misleading confidence in trends



*Error correlation matrix from
Giering et al. 2019*



Error Covariance



What is Error Covariance?

Combines error correlation and uncertainty

$$S = U R U^T$$



Random, systematic and structured uncertainty

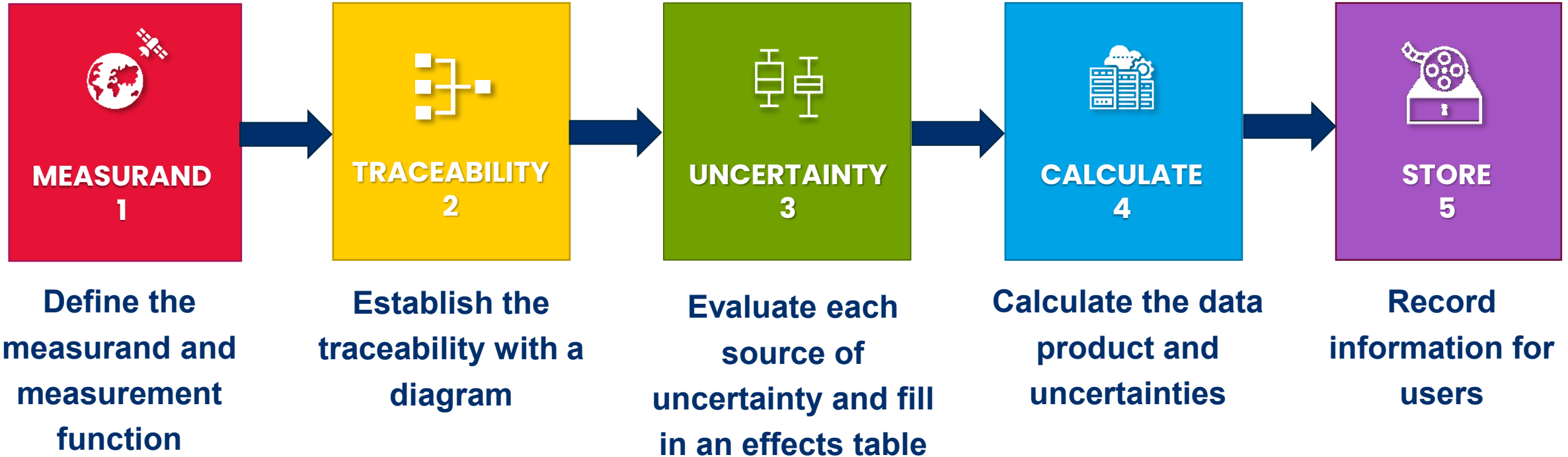
- ☐ It is the errors that are correlated, not the uncertainty values
- ☐ Random means completely uncorrelated, i.e. error correlation is identity matrix
- ☐ Systematic means fully correlated, i.e. error correlation filled with 1's



A Metrological Approach



Uncertainties are evaluated and expressed following [QA4EO Five Steps](#), a framework which employs the principles of metrology.



CoMet Toolkit



 **punpy**

Propagation UNcertainties in Python

 **obsarray**

Handling uncertainty and error-covariance in datasets

 **comet_maths**

CoMet mathematical algorithms and interpolation tools

 **UNC Specification**

Uncertainty metadata naming conventions

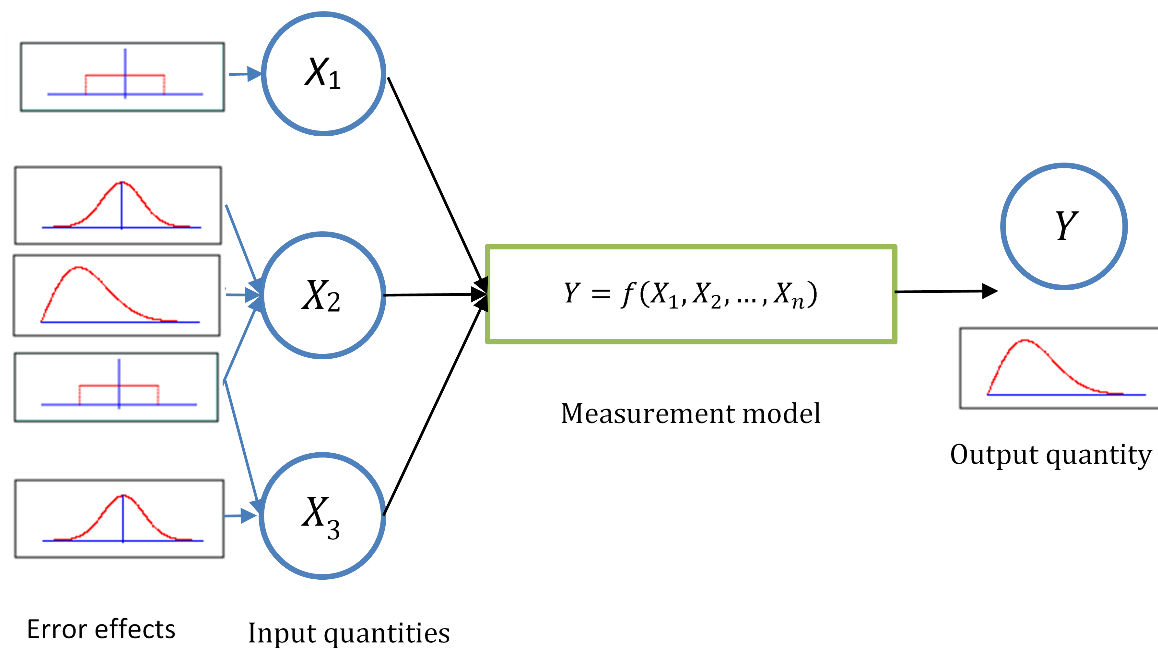
CoMet Packages:



 punpy

Propagating Uncertainties with punpy

- ❑ Python module for propagating **random**, **systematic** and **structured uncertainties** through any Python measurement function
- ❑ Flexible in terms of the **specified correlations** along given dimensions or between input quantities
- ❑ **Monte Carlo** and **Law of Propagation of uncertainties** methods available



Punpy as a Standalone Tool



□ Simple user **interface**:

1. Import punpy
2. Define measurement function
3. Create MC or LPU object
4. Propagate uncertainties

```
import punpy
prop=punpy.MCPropagation(10000)
unc_measurand=prop.propagate_random(measurement_func,
                                     [input_qt1,input_qt2],[unc_qt1,unc_qt2])
```

□ **Measurement function** are defined as python functions that take arrays as input quantities and return an array as measurand

□ Many optional **keywords** for flexible functionality

- *return_corr*
- *Corr_between*
- *Repeat_dims*
- *Parallel_cores*
- *Output_vars*
- ...

Punpy with digital effects tables

- ❑ **punpy** interfaces with **obsarray** to make uncertainty propagation as efficient and easy to use as possible
- ❑ **propagate_ds()** function returns an **obsarray** dataset with combined random, systematic and structured uncertainties on measurand

```
from punpy import MeasurementFunction

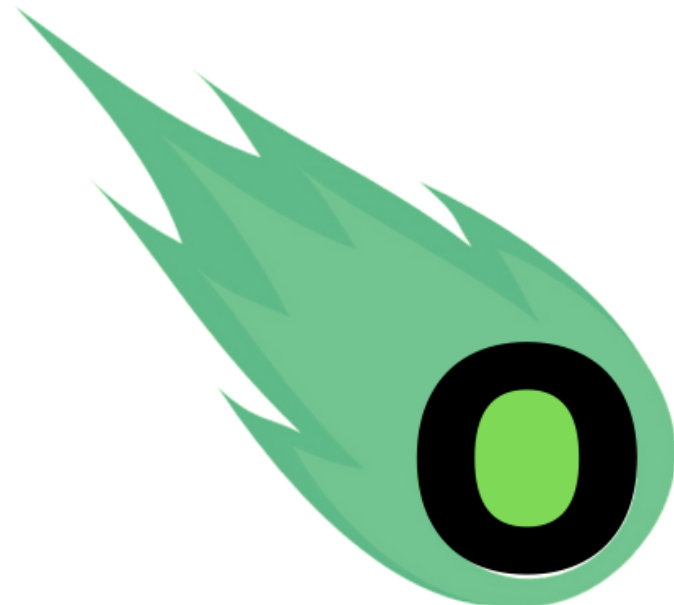
# Define your measurement function inside a subclass of MeasurementFunction
class IdealGasLaw(MeasurementFunction):
    def meas_function(self, pres, temp, n):
        return (n * temp * 8.134) / pres

# create object of the measurement function class and specify the variable names
gl = IdealGasLaw(["pressure", "temperature", "n_moles"], "volume", yunit="m^3")

# propagate uncertainties on the input quantities in ds to measurand in ds_y
ds_y = gl.propagate_ds(ds)
```

 CoMet Packages:



 obsarray



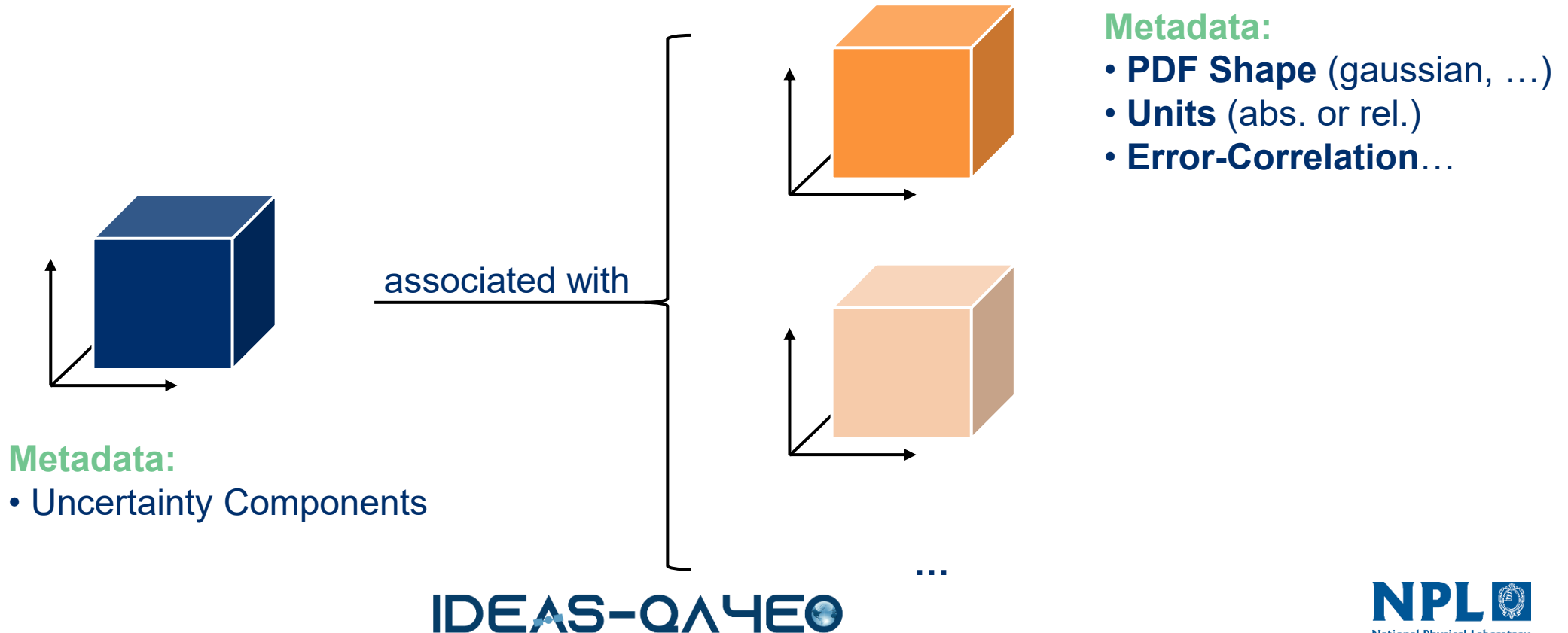
UNC Specification



Uncertainty Variable Metadata

Observation Variables

Uncertainty Variables



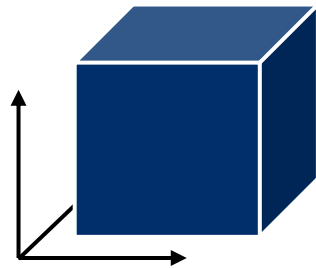


UNC Specification



Example – Temperature Dataset

Observation Variables



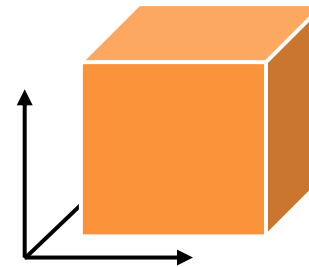
temperature

Metadata:

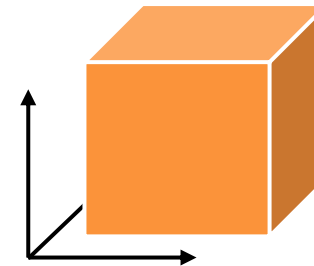
- Uncertainty Components –
[“u_noise”, “u_calibration”]

associated with

Uncertainty Variables



u_noise



u_calibration

Metadata:

- PDF Shape – “gaussian”
- Units – %
- Error-Correlation:
 - All dims - Random

Metadata:

- PDF Shape – “rectangular”
- Units – “K”
- Error-Correlation:
 - **x, y** – systematic
 - **time** – defined by matrix



UNC Specification



Example – Temperature Dataset

```
variables:
  float temperature(time, lat, lon);
    temperature:unc_comps=["u_calibration", "u_noise"];
    temperature:units="K"
  float u_calibration(time, lat, lon);
    u_calibration:units="K";
    u_calibration:pdf_shape="rectangular";
    u_calibration:err_corr_dim1_name=["lat", "lon"];
    u_calibration:err_corr_dim1_form="systematic";
    u_calibration:err_corr_dim2_name="time";
    u_calibration:err_corr_dim2_form="err_corr_matrix";
    u_calibration:err_corr_dim2_params=["err_corr_calibration_time"];
  float u_noise(time, lat, lon);
    u_calibration:err_corr_dim1_name=["time", "lat", "lon"];
    u_calibration:err_corr_dim1_form="random";
  float err_corr_calibration_time(time, time);
```



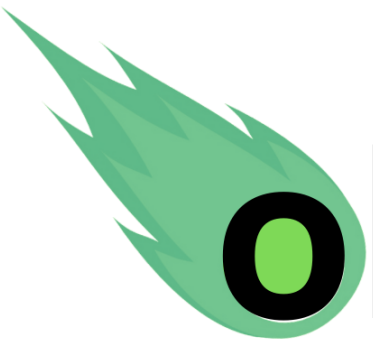
Measurement data handling in Python

- ❑ **obsarray** is an extension to xarray to support defining, storing and interfacing with measurement data – using the UNC specification.
- ❑ Also has functionality for defining flags following **CF Conventions**.
- ❑ Working towards including UNC spec within **CF Conventions**
- ❑ It is designed to work well with netCDF files and for the **Earth Observation** community.

Plugs straight into punpy for propagation through measurement functions!

Comet Application Examples

 punpy

 obsarray

 comet_maths

CoMet Toolkit in Action



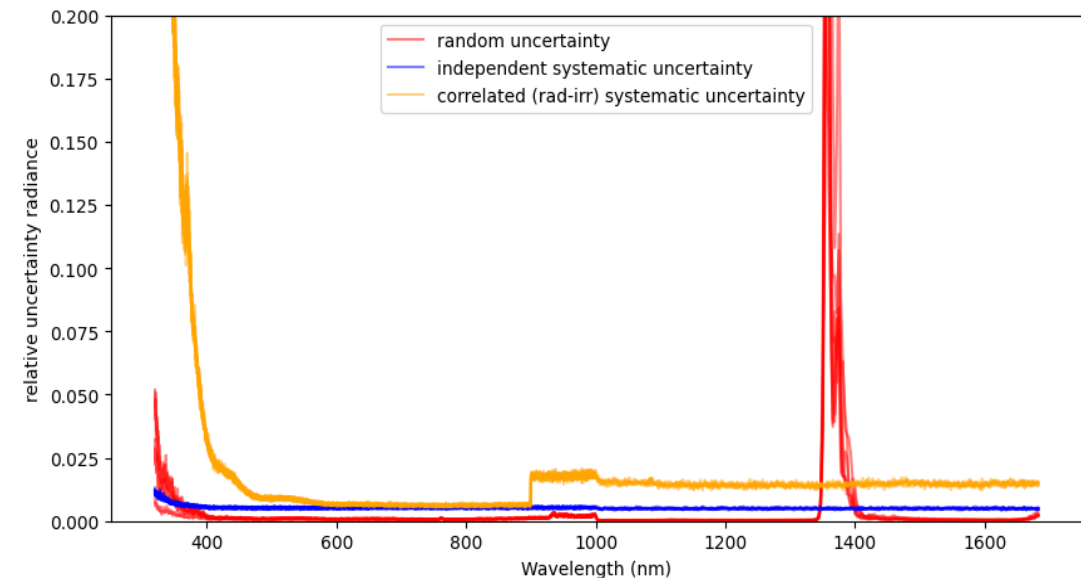
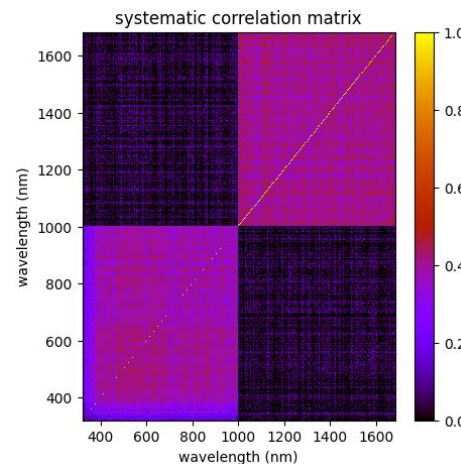
- ❑ Validated against **NIST** uncertainty engine:

https://colab.research.google.com/github/comet-toolkit/comet_training/blob/main/NIST_example.ipynb

- ❑ *CoMet* is used in various other projects, such as **QA4EO**, **HYPERNETS**, **CHIME L2**, **FLEX** validation, **TRUTHS** science studies, **LIME**, **FRM4SOC**, **RPV4PICS**
see <https://www.comet-toolkit.org/projects/>

- ❑ **Example** from *hypernets_processor* :

Hypernets is an automated network of in-situ instruments measuring reflectance for L2 satellite validation





Exercises



□ Please go to www.comet-toolkit.org/user-guide/training/



Today's Exercises



*comet-toolkit.org/user-
guide/training/vhroda*

IDEAS-QA4EO



CoMet Release



❑ V1.0 of Comet toolkit has been released as **open source** toolkit:

- www.comet-toolkit.org
- github.com/comet-toolkit



❑ Accompanied by training material (**Jupyter** notebooks hosted on google colab):

- www.comet-toolkit.org/examples

❑ Documentation & ATBD for individual tools:

- obsarray.readthedocs.io/en/latest/
- punpy.readthedocs.io/en/latest/
- comet-maths.readthedocs.io/en/latest/



Outlook



- ❑ **Current release** will be presented De Vis & Hunt (in prep)
- ❑ Looking to continue to expand the use cases the developed tools
 - Aiming to enable uncertainty propagation through **any python measurement function**
 -  Please get in touch if you are interested!
- ❑ This has been our first step into this way of working, **many more ideas in a roadmap**
 - MetEOR (e.g. CEOS-PVP Cal/Val pipeline, BRDF tool, ...)
- ❑ **1.5 day training course** tentatively planned for May at NPL



www.comet-toolkit.org/meteor/

<https://qa4eo.org/met4eo/>



Summary



- ❑ The **CoMet toolkit** is an open-source software project to develop Python tools for the handling of error-covariance information in the analysis of measurement data
- ❑ This toolkit is based on **robust metrology**, and makes dealing with complexities of uncertainties much easier
- ❑ Includes **obsarray**, **punpy** & **comet_maths** as initial offering, to be extended
- ❑ These tools are already being used operationally in various projects (e.g. Hypernets)